

CS173: Discrete Structures

Problem Set 9

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Problem 1. Show that, for every tree T , there exists a unique path between any two vertices in T .

Proof. Let T be a tree and let $v, w \in V(T)$.

We proceed by cases on $|V(T)|$. Since T is a tree, T is a finite graph, so $|V(T)| \in \mathbb{N}$. We consider $|V(T)| = 0$, $|V(T)| = 1$, $|V(T)| = 2$, and $|V(T)| \geq 3$ separately.

If $|V(T)| = 0$, then $V(T) = \emptyset$, so there are no vertices $v, w \in V(T)$. The claim $(\forall v, w \in V(T))(\text{there is a unique path from } v \text{ to } w)$ is vacuously true.

If $|V(T)| = 1$, then $v = w$ and the unique path is the trivial path $p : 1 \rightarrow V(T)$ with $p(0) = v$.

If $|V(T)| = 2$, then $V(T) = \{v, w\}$ and since T is connected, $\{v, w\} \in E(T)$. The only path is $p : 2 \rightarrow V(T)$ with $p(0) = v$ and $p(1) = w$ (or $v = w$ and the trivial path), so uniqueness holds.

Now suppose $|V(T)| \geq 3$. Since T is a tree, T is connected, so by definition $(\exists k \in \mathbb{N})(\exists p : (k+1) \rightarrow V(T))(p \text{ is a path with } p(0) = v \text{ and } p(k) = w)$.

Suppose for contradiction there exist two distinct paths from v to w : let $k_1, k_2 \in \mathbb{N}$ and let $p_1 : (k_1+1) \rightarrow V(T)$ and $p_2 : (k_2+1) \rightarrow V(T)$ be paths with $p_1(0) = p_2(0) = v$ and $p_1(k_1) = p_2(k_2) = w$ and $p_1 \neq p_2$.

Since $p_1 \neq p_2$, there exists a smallest $i \in \mathbb{N}$ such that $p_1(i+1) \neq p_2(i+1)$ (with $p_1(i) = p_2(i)$). Let $u := p_1(i)$.

Similarly, let $j_1, j_2 \in \mathbb{N}$ with $j_1 > i$ and $j_2 > i$ be the smallest indices such that $p_1(j_1) = p_2(j_2)$; such indices exist because at least $p_1(k_1) = p_2(k_2) = w$. Let $u' := p_1(j_1) = p_2(j_2)$.

Define $c : ((j_1 - i) + (j_2 - i) + 1) \rightarrow V(T)$ by following p_1 forward from u to u' , then p_2 backward from u' to u :

$$c(m) := \begin{cases} p_1(i+m) & \text{if } 0 \leq m \leq j_1 - i, \\ p_2(j_2 - (m - (j_1 - i))) & \text{if } j_1 - i < m \leq (j_1 - i) + (j_2 - i). \end{cases}$$

Then $c(0) = p_1(i) = u$ and $c((j_1 - i) + (j_2 - i)) = p_2(i) = u$, so c is a circuit. By the minimality of i, j_1 , and j_2 , no internal vertex repeats, so c is a cycle. But T is a tree, so T contains no cycles — a contradiction.

Therefore the path from v to w is unique. □

Problem 2. Prove that every tree T satisfying $|V(T)| \geq 2$ must contain at least one leaf node.

Proof. Let T be a tree with $|V(T)| \geq 2$. Since T is connected and $|V(T)| \geq 2$, there is at least one edge, so there exists a path of length ≥ 1 .

Every path $q : (m + 1) \rightarrow V(T)$ (for any $m \in \mathbb{N}$) is injective by definition, so q visits $m + 1$ distinct vertices, each belonging to $V(T)$. Hence $m + 1 \leq |V(T)|$, i.e., $m \leq |V(T)| - 1$. Since the length of any path in T is bounded above by $|V(T)| - 1$, a path of maximum length exists. Let $k \in \mathbb{N}$ with $k \geq 1$ and let $p : (k + 1) \rightarrow V(T)$ be such a path.

Define the set $S := \{y \in V(T) \mid (\exists x \in (k+1))(p(x) = y)\} = \{p(0), p(1), \dots, p(k)\}$.

Let $v := p(k)$ (the last vertex of p). We claim $\deg_T(v) = 1$, i.e., v is a leaf.

Since $\{p(k-1), v\} \in E(T)$, we have $\deg_T(v) \geq 1$.

Suppose for contradiction that $\deg_T(v) \geq 2$. Since $\{p(k-1), v\} \in E(T)$ accounts for one incident edge, $(\exists w \in N_T(v))(w \neq p(k-1))$. Fix such a w .

Case 1: $w \notin S$. Define $p' : (k + 2) \rightarrow V(T)$ by $p'(i) := p(i)$ for $0 \leq i \leq k$ and $p'(k + 1) := w$. Since $w \notin S$, w is distinct from every vertex in S , so p' is injective and hence p' is a path of length $k + 1$, contradicting the maximality of p .

Case 2: $w \in S$. Then $(\exists j \in \mathbb{N})(0 \leq j \leq k \text{ and } w = p(j))$. Fix such a j . Since $\{w, v\} \in E(T)$ and graphs have no self-loops, $w \neq v = p(k)$, so $j \neq k$. Since $w \neq p(k-1)$ by choice of w , we have $j \leq k - 2$.

Define the walk $c : (k - j + 2) \rightarrow V(T)$ by $c(i) := p(j + i)$ for $0 \leq i \leq k - j$ and $c(k - j + 1) := p(j)$. Then $c(0) = c(k - j + 1) = p(j)$, consecutive vertices are connected by edges of p , the last edge $\{c(k - j), c(k - j + 1)\} = \{p(k), p(j)\} = \{v, w\} \in E(T)$, and no vertex repeats except $p(j)$ at the start and end (by injectivity of p). So c is a cycle in T , but T is a tree, so T contains no cycles — a contradiction.

Both cases yield a contradiction, so $\deg_T(v) = 1$ and v is a leaf. $\square$

Problem 3. Use induction to prove that every nonempty tree T satisfies $|V(T)| = |E(T)| + 1$.

Proof. Let $P(n)$ be the predicate

$$(\forall T)((T \text{ is a tree} \wedge T \text{ is nonempty} \wedge |V(T)| = n) \Rightarrow (|V(T)| = |E(T)| + 1)).$$

We prove $(\forall n \in \mathbb{N})(n \geq 1 \Rightarrow P(n))$ by induction on n .

Base case. We show $P(1)$. Let T be a nonempty tree with $|V(T)| = 1$. Then $V(T) = \{v\}$ for some vertex v . Since graphs have no self-loops, no edge in $E(T)$ can have v as both endpoints, and v is the only vertex, so $E(T) = \emptyset$. Hence $|E(T)| = 0$ and $|V(T)| = 1 = 0 + 1 = |E(T)| + 1$. So $P(1)$ holds.

Inductive step. Let $n \in \mathbb{N}$ with $n \geq 1$ and assume $P(n)$, i.e.,

$$(\forall T')((T' \text{ is a tree} \wedge T' \text{ is nonempty} \wedge |V(T')| = n) \Rightarrow (|V(T')| = |E(T')| + 1)).$$

We show $P(n + 1)$.

Let T be a nonempty tree with $|V(T)| = n + 1$. Since $n \geq 1$, we have $|V(T)| \geq 2$, so by Problem 2, $(\exists \ell \in V(T))(\deg_T(\ell) = 1)$. Fix such a leaf ℓ . Since $\deg_T(\ell) = 1$, there is exactly one edge incident on ℓ ; let e_ℓ be this edge, i.e., $\{e_\ell\} = \{e \in E(T) \mid \ell \in e\}$.

Define the graph T' by $V(T') := V(T) \setminus \{\ell\}$ and $E(T') := E(T) \setminus \{e_\ell\}$. Then $|V(T')| = n$ and $|E(T')| = |E(T)| - 1$.

We verify T' is a nonempty tree:

Nonempty. $|V(T')| = n \geq 1$, so $V(T') \neq \emptyset$.

Connected. $(\forall u, w \in V(T'))$ there exists a path from u to w in T (since T is connected). This path does not pass through ℓ , because ℓ is a leaf: if the path entered ℓ , it would have to exit through the same edge (the only one incident on ℓ), revisiting the neighbor of ℓ , contradicting injectivity. Hence the path lies entirely in T' , so T' is connected.

No cycles. Any cycle in T' would also be a cycle in T (since $E(T') \subseteq E(T)$), contradicting T being a tree. So T' contains no cycles.

So T' is a nonempty tree $\wedge |V(T')| = n$, so

$$\begin{aligned} |V(T')| &= |E(T')| + 1 && \text{(by } P(n)) \\ |V(T)| - 1 &= (|E(T)| - 1) + 1 && \text{(by single-element removal)} \\ |V(T)| &= |E(T)| + 1 \end{aligned}$$

Therefore $P(n + 1)$ holds, and by induction $P(n)$ holds for all $n \in \mathbb{N}$ with $n \geq 1$. $\square$

Problem 4. Prove any graph G satisfying $(\forall v \in V(G))(\deg_G(v) \geq \frac{|V(G)|-1}{2})$ is connected.

Proof. Let G be a graph with $|V(G)| = n \in \mathbb{N}$ and $(\forall v \in V(G))(\deg_G(v) \geq \frac{n-1}{2})$.

Let $u, w \in V(G)$ be arbitrary distinct vertices. We show there is a path from u to w .

Case 1: $\{u, w\} \in E(G)$. Then the function $p : 2 \rightarrow V(G)$ given by $p(0) = u$, $p(1) = w$ is a path of length 1 from u to w .

Case 2: $\{u, w\} \notin E(G)$. We show $(\exists x \in V(G))(x \in N_G(u) \text{ and } x \in N_G(w))$.

Since $w \notin N_G(u)$ and $u \notin N_G(w)$, we have $N_G(u) \subseteq V(G) \setminus \{u, w\}$ and $N_G(w) \subseteq V(G) \setminus \{u, w\}$, so both neighborhoods are subsets of a set of size $n - 2$.

By the degree condition, $|N_G(u)| \geq \frac{n-1}{2}$ and $|N_G(w)| \geq \frac{n-1}{2}$.

Suppose for contradiction $N_G(u) \cap N_G(w) = \emptyset$. Then

$$n - 2 \geq |N_G(u) \cup N_G(w)| = |N_G(u)| + |N_G(w)| \geq \frac{n-1}{2} + \frac{n-1}{2} = n - 1.$$

This gives $n - 2 \geq n - 1$, a contradiction.

Hence $(\exists x \in N_G(u) \cap N_G(w))$. Fix such an x . Then $\{u, x\} \in E(G)$ and $\{x, w\} \in E(G)$, so the function $p : 3 \rightarrow V(G)$ defined by $p(0) = u$, $p(1) = x$, $p(2) = w$ is a path of length 2 from u to w .

In both cases, there exists a path from u to w , so G is connected. □